25901545 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSTA200 - BASICS OF MULTIVARIATE DISTRIBUTIONS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. Why do we obtain two regression lines when analyzing the relationship between two variables? Explain. [An] / [CO1]
2. Explain Kendall's Tau coefficient. [An] / [CO1]
3. Prove that $\mathrm{Cov}(aX, bY) = ab\ \mathrm{Cov}(X, Y)$, using mathematical expectation properties. [A] / [CO2]
4. State and prove the multiplication theorem of mathematical expectation. [U] / [CO2]
5. Give two characterizations of multivariate normal distribution. [K] / [CO3]
6. State any two applications of multinomial distribution. [K] / [CO3]
7. Write the likelihood function of the sample for a MVN Distribution. [A] / [CO3]
8. Let X~N(μ,Σ) Find the mean and variance of l'x, where l is a vector of scalars. [A] / [CO3]
9. What is the marginal distribution of $X^{(1)}$ of 'q' random variables of $X=(X^{(1)},X^{(2)}) \sim Np(\mu, \sum)$. [U] / [CO3]

10. Write the formula for Var(X'AX) when X follows $N_p(0, \Sigma)$. [A] / [CO4]

[2x7 = 14]

Part B

Short essay type questions

Answer any **4** questions

Each question carries **6** marks

11. State and prove the invariance property of the correlation coefficient. [An] / [CO1]

12. Explain the derivation of Spearman's rank correlation coefficient formula. [An] / [CO1]

13. $f(x,y) = \frac{1}{72}(2x+3y), \quad x = 0,1,2, \quad y = 1,2,3$ is the joint p.d.f of (X,Y). [An] / [CO2]

    1. Find the distribution of $X+Y$
    2. Find the conditional distribution of X given $X+Y=3$
    3. Examine whether X and Y are independent.

14. Determine k such that the joint distribution of a pair (X,Y) of continuous random variables is $f(x,y) = k(xy+2x+3y+6), \quad 0 \le x,y \le 1$. Examine whether X and Y are independent. [A] / [CO2]

15. Derive the necessary and sufficient condition for the independence of two sub-vectors of a normal vector. [A] / [CO3]

16. State and prove the conditions for the two quadratic forms X'AX and X'BX are independent if X follows $N_p(0,\Sigma)$, [U] / [CO4]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. (a) Define a scatter diagram and explain its role in analyzing relationships between variables. [A] / [CO1]

    (b) Construct a scatter diagram for the following data

x	1	3	5	7	8	10
y	8	12	15	17	18	20

18. Show that X'AX follows chi square distribution with r degrees of freedom and A is idempotent if X follows $N_p(0, I)$. [U] / [CO4]

[12x1 = 12]